

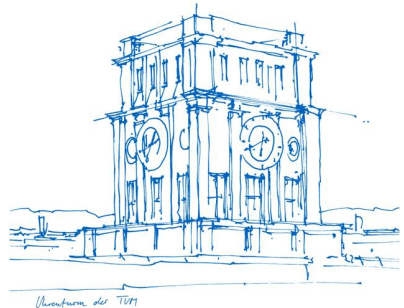
Graph-Based Compression of Agglomerate Morphology

Predicting geometric exposure from DEM-resolved structures

Preliminary work toward morphology-aware
oxygen-transport closures

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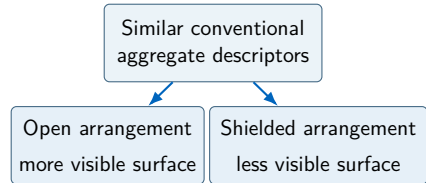
Internal arrangement can change oxygen access

Two agglomerates may have similar

- particle number and overall size;
- porosity or fractal dimension;
- primary-particle properties.

Yet they may differ in

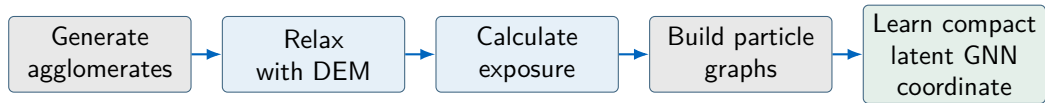
- particle shielding;
- surface accessibility;
- internal transport pathways.



A useful morphology descriptor must distinguish structures that integral scalars treat as similar.

Testing the structural workflow before oxygen simulations

Can a particle-contact graph be compressed without losing the information needed to predict geometric exposure?



Prediction

How accurately can the graph predict exposure?

Compression

How many latent variables are needed?

Interpretation

What structural information is retained?

The data set separates size, morphology and random realization

$$N = k_f \left(\frac{R_g}{r_p} \right)^{D_f},$$

$3 d_p \times 3 N \times 3 \text{ morphologies} \times 5 \text{ realizations} = 135$.

N : particle count; R_g : radius of gyration;
 r_p : particle radius;
 D_f : requested fractal dimension;
 k_f : requested fractal prefactor.

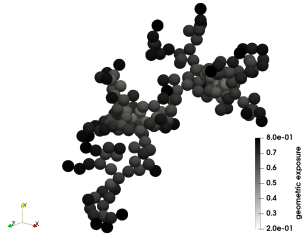
Morphology	D_f	k_f
Open	1.8	1.3
Intermediate	2.2	1.1
Compact	2.6	0.8

Each design point contains five independently generated particle arrangements.

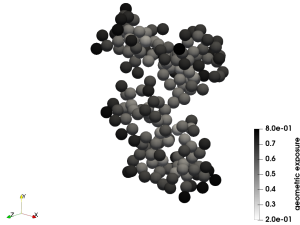
The three morphology classes use paired D_f and k_f settings; their individual effects cannot be separated here.

Agglomerate generation: Filippov et al. (2000); hierarchical implementation: Skorupski et al. (2014).

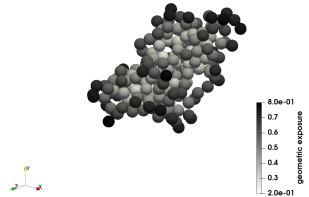
The ensemble contains visibly different particle arrangements



Open
 $(D_f, k_f) = (1.8, 1.3)$



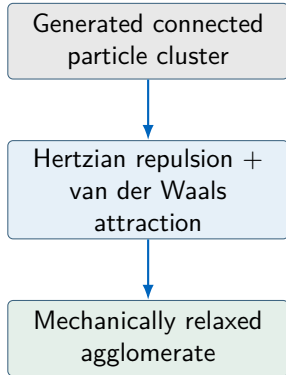
Intermediate
 $(D_f, k_f) = (2.2, 1.1)$



Comparatively compact
 $(D_f, k_f) = (2.6, 0.8)$

**All three examples have $N = 200$ and $d_p = 2 \mu\text{m}$;
 colour denotes particle-level exposure from 0.2 to 0.8**

DEM relaxation turns generated geometry into a physical structure



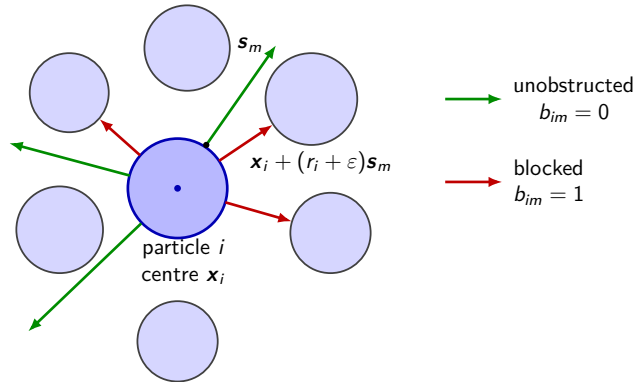
Why relax the structures?

- Remove small overlaps and force imbalance from generation.
- Test whether the cluster remains mechanically connected.
- Obtain the final contact geometry used by the graph.

Quality checks

- low residual kinetic energy;
- one connected component;
- consistent particle and contact output.

Ray tracing measures unobstructed particle directions



The calculation uses 512 approximately uniform directions over the full sphere; only a 2D subset is shown.

Exposure is a number between zero and one

$$a_i^{\text{geo}} = \frac{1}{M} \sum_{m=1}^M (1 - b_{im}),$$
$$\bar{a}_{\text{geo}} = \frac{1}{N} \sum_{i=1}^N a_i^{\text{geo}}.$$

$M = 512$: sampled ray directions;
 b_{im} : obstruction indicator for ray m of particle i ;
 N : number of particles.

Particle-level meaning

$a_i^{\text{geo}} = 0$: every sampled direction is blocked.

$a_i^{\text{geo}} = 1$: every sampled direction is unobstructed.

Aggregate target

$\bar{a}_{\text{geo}}$ is the mean exposure over all particles.

Exposure is a geometric line-of-sight proxy—not yet oxygen diffusion, uptake or reaction.

Why predict geometric exposure?

Closer to the uptake mechanism

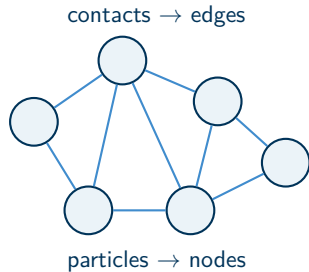
- Geometric exposure describes the particle surface that is potentially accessible from the surroundings.
- It is therefore directly relevant to possible oxygen uptake.
- Porosity and fractal dimension describe the overall structure, but not the accessible surface directly.
- An additional model is needed to connect them to oxygen uptake.

Better suited to the present data set

- The data set contains 135 agglomerates but only three prescribed fractal dimension classes ($D_f = 1.8, 2.2, 2.6$)
- This is insufficient for learning a general relationship between particle structure and fractal dimension.
- Geometric exposure varies continuously among the 135 agglomerates.
- It also distinguishes random realizations belonging to the same morphology class.

Geometric exposure is a continuous and uptake-relevant target; porosity and D_f remain useful structural descriptors.

The relaxed agglomerate becomes a particle-contact graph



Node information

- radial position: $|\mathbf{x}_i - \mathbf{x}_c|/R_g$;
- coordination number: z_i .

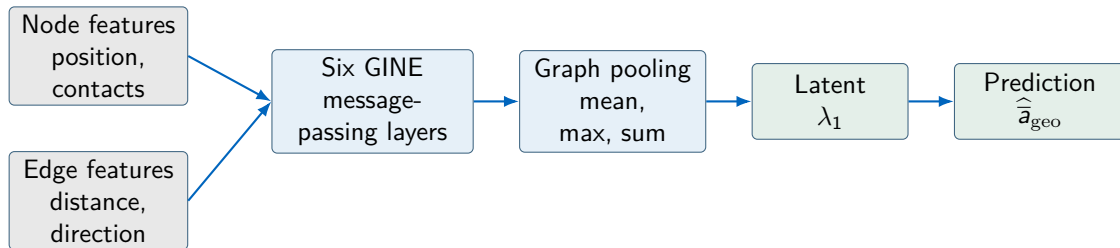
Edge information

- normalized centre distance: d_{ij}/d_{ref} ;
- normalized relative-position vector.

Only geometrically touching or numerically near-touching pairs are retained.

The graph preserves local neighbourhoods and the arrangement of particles across the complete agglomerate.

The GNN compresses the whole graph into one learned coordinate



Messages

combine each particle with its neighbours.

Pooling

produces one representation per agglomerate.

Bottleneck

forces the useful information into λ_1 .

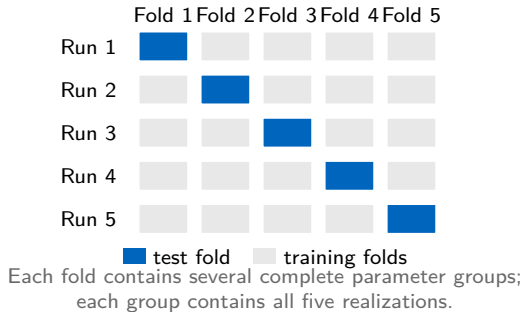
No separately calculated aggregate descriptor is supplied to the latent coordinate or prediction head.

Grouped cross-validation tests unseen parameter combinations

Grouping rule

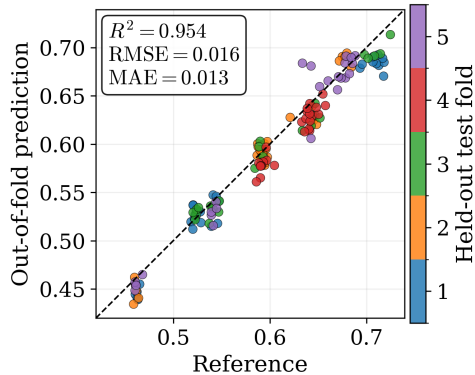
- The 135 agglomerates form 27 parameter groups; each group contains five realizations with the same (d_p, N, D_f, k_f) setting.
- The 27 complete groups are distributed among five folds.
- In each run, one fold is tested and the other four are used for training. A group is never divided between them.

27 groups divided into five folds



Every reported prediction is out-of-fold: it comes from a model that did not train on that parameter group.

One graph coordinate predicts exposure with $R^2 = 0.954$



Grouped out-of-fold performance

$$\text{RMSE} = 0.0160$$

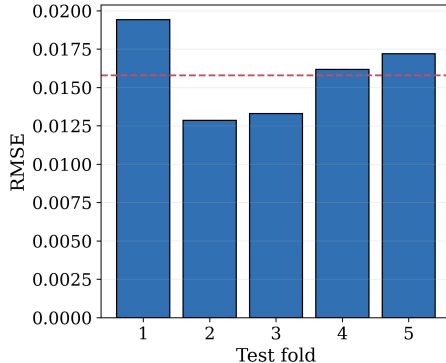
$$\text{MAE} = 0.0128$$

$$R^2 = 0.954$$

Points close to the diagonal indicate agreement between calculated and GNN-predicted mean exposure.

A one-dimensional bottleneck retains nearly all information needed for this specific prediction task.

Prediction accuracy remains consistent across the five test folds



Fold-specific RMSE

- Fold 1: 0.0194
- Fold 2: 0.0128
- Fold 3: 0.0133
- Fold 4: 0.0162
- Fold 5: 0.0172

The dashed line in the figure denotes the mean fold RMSE.

The performance is not produced by one unusually easy split.

The particle graph adds information beyond conventional scalars

Model or input	RMSE	MAE	R^2
Pure-graph GNN	0.0160	0.0128	0.954
Six calculated descriptors, linear	0.0209	0.0183	0.922
Six calculated descriptors, random forest	0.0235	0.0168	0.902
Requested D_f and k_f , linear	0.0472	0.0399	0.604
Best individual descriptor	0.0524	0.0383	0.511
Particle count, linear	0.0687	0.0582	0.160

$D_f + k_f$ are informative

but they describe only the prescribed morphology class.

The graph is more precise

because it sees realization-, size- and relaxation-dependent structure.

The learned coordinate is task-specific, not a renamed scalar

Strongest monotonic associations

Descriptor	Spearman ρ
Requested D_f	-0.800
Requested k_f	+0.800
Number of contacts	-0.596
Particle count	-0.544
Maximum coordination	-0.541

- One scalar alone does not reconstruct the learned coordinate well.
- Several morphology measures together reconstruct it accurately.
- Its sign is arbitrary; its encoded ordering and predictive content matter.

λ_1 combines several structural attributes in the direction most useful for predicting exposure.

The present result is a structural proof of concept

What has been demonstrated

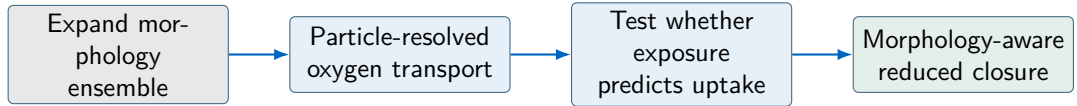
- stable DEM-to-graph workflow for 135 agglomerates;
- physically interpretable ray-based exposure target;
- accurate exposure prediction with one latent coordinate;
- improvement over prescribed and calculated scalar descriptors.

What has not yet been demonstrated

- oxygen diffusion through tortuous pores;
- flow-induced transport around or within an agglomerate;
- oxygen consumption and reaction–transport coupling;
- direct prediction of biological activity.

Geometric exposure is a useful screening target, but its relationship to oxygen uptake must be validated.

Resolved oxygen transport is the decisive next test



Geometry checks

ray-number convergence and
finite probe radius

Transport targets

uptake, response time and
oxygen-limited fraction

Model decision

retain exposure or enrich the
graph with transport
information

The detailed graph provides a compact bridge to transport models

A single learned graph coordinate predicts ray-based geometric exposure with grouped out-of-fold $R^2 = 0.954$.

Meaningful target

Exposure quantifies directional surface obstruction from 0 to 1.

Useful compression

One latent coordinate retains task-relevant graph information.

Clear next step

Validate the structural proxy against resolved oxygen transport.

The proof of concept supports using graph-based morphology descriptors in the planned oxygen-transport study.